

Intra-Urban Heat Island Effect Analysis Through Geospatial Techniques: A Case Study Of Chandigarh

*A project report submitted in partial fulfilment of the
requirements for the degree of*

Bachelor of Technology

Submitted by

Kartikeya Gaur (20bce004)

Lakshit Juneja (20bce006)

Tarun Agarwal (20bce042)

Utkarsh Kumar (20bce080)

Under the guidance of

Dr. Chander Prakash



Department of Civil Engineering

National Institute of Technology Hamirpur

Hamirpur, India-177005

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राष्ट्रीय प्रौद्योगिकी संस्थान हमीरपुर

हमीरपुर (हि.प्र.) – 177 005 (भारत)

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NATIONAL INSTITUTE OF TECHNOLOGY HAMIRPUR

HAMIRPUR (H.P.) - 177 005 (INDIA)

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We hereby certify that the work which is being presented in the project report titled **Intra-Urban Heat Island Effect Analysis Through Geospatial Techniques: A Case Study Of Chandigarh** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology and submitted in the **Civil Engineering Department**, National Institute of Technology Hamirpur, is an authentic record of our own work carried out during a period from “August 2023 to May 2024” under the guidance of **Dr. Chander Prakash**, Associate Professor in the Department of Civil Engineering at the National Institute of Technology, Hamirpur.

The matter presented in this project report has not been submitted by me for the award of any other degree of this or any other Institute/University.

Kartikeya Gaur (20bce004)

Lakshit Juneja (20bce006)

Tarun Agarwal (20bce042)

Utkarsh Kumar (20bce080)

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Dr. Chander Prakash
(Associate Professor)

The Project Viva-Voce Examination has been held on

Dr. Chander Prakash
(Associate Professor)

Dr. V.K. Bansal
(Head of Department)

Signature of External Examiner(s)

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Kartikeya Gaur (20bce004)

Lakshit Juneja (20bce006)

Tarun Agarwal (20bce042)

Utkarsh Kumar (20bce080)

ABSTRACT

Rapid urbanization and human activities are major contributors to the formation of urban heat islands (UHIs), causing a range of environmental challenges. This study explores the spatial variations of UHI effects in Chandigarh, a planned city in India.

We leveraging Land Surface Temperature (LST) data acquired from the Thermal Bands of USGS Landsat 8 satellite over the past two decades (2003-2023). This data allows us to analyze both the spatial and temporal variations in Chandigarh's thermal environment. Our analysis identifies zones experiencing different UHI intensities and explores the distribution of these zones across the city.

The results reveal a significant UHI effect in Chandigarh, with an average temperature increase of 2.798°C over the past 20 years, translating to an average annual increase of 0.1399°C . In comparison to unplanned cities like Jalandhar, which witnessed a rise of 6.77°C (average annual increase of 0.3385°C) over the same period, Chandigarh's planned layout appears to offer some mitigation against the UHI effect.

This research highlights the importance of understanding UHI variations within cities. By identifying areas experiencing the most intense heat through geostatistical analysis marking significant areas impacting the public health. These strategies may include promoting green spaces, urban forestry programs, and energy-efficient building practices.

This study underlines the need for a comprehensive approach to urban development that balances the city's growth with the well-being of its residents.

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List of Acronyms

UHI	Urban Heat Island
ALOS	Advanced Land Observing Satellite
LST	Land Surface Temperature
CSV	Comma-separated values
DEM	Digital Elevation Model
GIS	Geographic information system
LULC	Land Use Land Cover
NDWI	Normalized Difference Water Index
SWIR	Short Wave Infrared
OLI	Operational Land Imager
TIRS	Thermal Infrared Sensors

Chapter 1

Introduction

Urbanization is a relatively new phenomenon in human history. Only 10% of the world's population settled in urban areas during the early 20th century; this percentage went up to 31% at the start of the 21st century; and now, in the 21st century, more than 62% of the world's population is living in urban areas (Ritchie and Roser, 2018). This drastic process is brought about by converting natural ground surfaces, such as bare soil, vegetation, water bodies, etc., to built-up areas mainly composed of asphalt, bitumen, concrete, rock, brick, etc., without much regard to its aftereffects. These materials absorb a large amount of solar radiation and retain it for longer periods of time, increasing the LST of the area (Mathew et al., 2016). Moreover, the loss of vegetation, natural habitat, etc. adversely affects the ecosystem. Although our cities expand spatially and seem to take up a lot of land, only around 2% of the total global land area is covered by built-up areas (Ritchie and Roser, 2018). The concentration of built-up areas and populations in limited spaces puts additional strain on resources such as land, water, mineral deposits, and so on, resulting in centralized use of these resources, the formation of microclimates, and changes in the eco- system over these regions (Tong et al., 2017). The changes in the natural habitat and the increase in anthropogenic activities in a limited space (cities) lead to a rise in temperature in the cities compared to the rural areas, known as the UHI effect (Voogt and Oke, 2003; Mathew et al., 2022).

Rapid urbanization presents a significant challenge for cities worldwide, characterized by a conversion of natural landscapes with low albedo (reflectivity) like bare soil and vegetation (albedo around 0.2) to built-up areas dominated by materials with high thermal admittance (heat absorption) such as asphalt (albedo around 0.09) and concrete (albedo around 0.17) (Mathew et al., 2016). This phenomenon triggers the Urban Heat Island (UHI) effect, causing cities to experience significantly higher Land Surface Temperature (LST) compared to surrounding rural areas. Studies have shown that cities can be 10-15°C hotter than their rural surroundings (European Commission, Joint Research Centre [JRC], n.d.). UHI not only impacts human health and comfort but also disrupts the natural energy balance, affecting air and water quality (Tong et al., 2017).

Traditionally, air temperature measured by weather stations has been used to quantify UHI (Li et al., 2018). However, the limited number of weather stations, particularly within dense urban environments, can lead to inaccurate assessments.

Several factors contribute to the intensity of UHI. Geographical location plays a role, with cities experiencing strong solar radiation more susceptible (Li et al., 2011). Urban morphology, characterized by dense building arrangements and limited green spaces, can trap heat and restrict natural ventilation, further amplifying UHI. For instance, a study in Baltimore found that the difference between 2-meter air temperatures in urban and rural areas could be as high as 3°C during heatwaves (Tian et al., 2021). Additionally, anthropogenic heat emissions from traffic and industrial activity significantly elevate LST within the city. Emissions from vehicles and industrial processes can increase local temperatures by several degrees (Lee et al., 2019). Meteorological factors like wind speed and direction (affecting heat dispersion), precipitation (contributing to evaporative cooling), and cloud cover (influencing solar radiation) also influence UHI by affecting the surface energy balance (Yang et al., 2020).

Remote sensing offers a valuable alternative. In our research, we specifically utilized Landsat thermal bands data to retrieve LST across the entire city, providing comprehensive spatial coverage of UHI (Derdouri et al., 2021). While LST and air temperature are not identical, they exhibit a strong correlation, enabling researchers to effectively identify and quantify UHI in cities (Mathew et al., 2016; Bala et al., 2020; Ramsay et al., 2023). Atmospheric UHI studies require more manpower, financial resources, materials, and time compared to LST-based approaches (Tian et al., 2021).

The consequences of UHI are severe. Elevated temperatures exacerbate heat stress for urban residents (Abdel-Ghany et al., 2013), potentially leading to heat-related illnesses and mortality (Arifwidodo et al., 2019; Goggins et al., 2012). UHI also disrupts natural atmospheric circulation patterns, contributing to air and water quality deterioration (Luo & Lau, 2018; Yao et al., 2019). Furthermore, agricultural practices on the outskirts of cities can be negatively impacted by the altered thermal regime caused by UHI (He et al., 2022; Ciardini et al., 2019). The increased reliance on air conditioning to mitigate discomfort further intensifies UHI through additional waste heat generation, creating a vicious cycle (Hwang et al., 2017; Du et al., 2016).

It's important to note that UHI is not uniformly distributed across a city. Densely built-up areas with minimal vegetation experience significantly higher LST compared to areas with parks, green spaces, and water bodies. These hotter pockets within the city are known as intra-city heat islands. Understanding these variations is crucial for developing targeted urban planning strategies.

While most UHI research has focused on temperate regions (Zhou et al., 2018; Ramsay et al., 2023), tropical cities face even greater challenges due to the compounding effect of high baseline temperatures and humidity (Siddiqui et al., 2021). Given the relentless pace of global urbanization (Ritchie & Roser, 2018), understanding UHI is essential for sustainable urban planning in all regions. By identifying vulnerable areas with high LST and limited natural cooling mechanisms, as revealed by Landsat thermal band data analysis, cities can develop resource-efficient plans that promote cool materials, green spaces, and strategic urban design. These measures can help create a more equitable distribution of heat within the city, improve public health, and enhance overall quality of life.

Identifying the most critical areas within a city disproportionately impacted by UHI is crucial for effective mitigation strategies. Here, hotspot analysis, a spatial statistical technique, comes into play. By analyzing LST data derived from Landsat satellites, hotspot analysis can pinpoint areas with statistically significant clusters of high temperatures. These hot spots represent the city's most vulnerable neighborhoods experiencing the most intense urban heat. By focusing on these areas for interventions like green space development and cool roof initiatives, urban planners can maximize the impact of their efforts and create a more equitable distribution of heat within the city.

Advances in remote sensing techniques and the increasing availability of data have enabled researchers to characterize UHI effects globally (Peng et al., 2012; Chakraborty and Lee, 2019). However, the focus has primarily been on temperate regions, such as North America, Europe, and China (Zhou et al., 2018; Ramsay et al., 2023). UHI studies in tropical regions, where UHIs compound the already high temperatures and humidity, are less common (Siddiqui et al., 2021). Given the rapid urbanization and population growth in India, there is a pressing need to investigate the impacts of UHIs on Indian cities. The current study examines the UHI effects in Chandigarh UT and identifies the areas that are most vulnerable to these effects.

Our study focuses on Chandigarh, a planned city in northwestern India. Chandigarh presents a unique case study for UHI research due to several factors. First, its rapid urbanization offers a clear example of how conversion of natural landscapes to built environments can exacerbate UHI. Second, Chandigarh's location in a region with intense summer sun makes it particularly susceptible to heat accumulation. Finally, the city's planned layout, with well-defined sectors and green belts, provides an opportunity to assess the effectiveness of urban design strategies in mitigating UHI. By analyzing LST data from Landsat satellites and incorporating land use information, our research aims to identify vulnerable areas within Chandigarh and evaluate how the city's unique morphology influences UHI patterns.

Chapter 2

Study Area

Chandigarh, a union territory and a meticulously planned city in northern India, serves as the shared capital for Punjab and Haryana, strategically positioned at the foothills of the Shivalik Hills. With an average elevation of 321 meters, spanning 114 square kilometers, and a perimeter of 105 kilometers, Chandigarh is renowned for its modern architecture and high quality of life. The city's rapid urbanization, fueled by economic growth, superior living conditions, and proximity to major cities, has propelled its population to 1,055,910 in 2011, expected to reach 1.6 million by 2031.

This surge in urbanization brings about challenges, including heightened demand for housing and strained infrastructure, leading to escalating traffic congestion and pollution. The Chandigarh administration is actively addressing these challenges with sustainable urban development initiatives. These efforts encompass the construction of new housing projects, substantial investments in public transportation to alleviate congestion, and comprehensive environmental measures aimed at improving air quality and water conservation. As Chandigarh grapples with the complex impacts of urbanization, its proactive and holistic approach positions it as a noteworthy model for other cities navigating similar challenges in India and planned city in northern India, serving as the shared capital of the surrounding states, namely Punjab to the north, west, and south, and Haryana to the east. It is located at the foothills of the Shivalik Hills, at an average elevation of 321 meters (1,053 feet) above sea level. The city covers an area of 114 square kilometers (44 square miles) and has a perimeter of approximately 105 kilometers (65 miles).

Situated in the northern part of India, Chandigarh experiences a humid subtropical climate, classified as Cwa under the Köppen climate classification system. The city experiences hot summers and mild winters, with average monthly temperatures ranging from 14.2°C (January) to 32.9°C (May). The annual average temperature in Chandigarh is 25.55°C.

Chandigarh receives an average annual rainfall of 1110 mm, with most of it falling during the monsoon season from June to September. The city is located at an elevation of 345 meters above sea level, nestled in the foothills of the Himalayas. Surrounded by hills on all sides, Chandigarh enjoys a relatively mild climate compared to other parts of India.

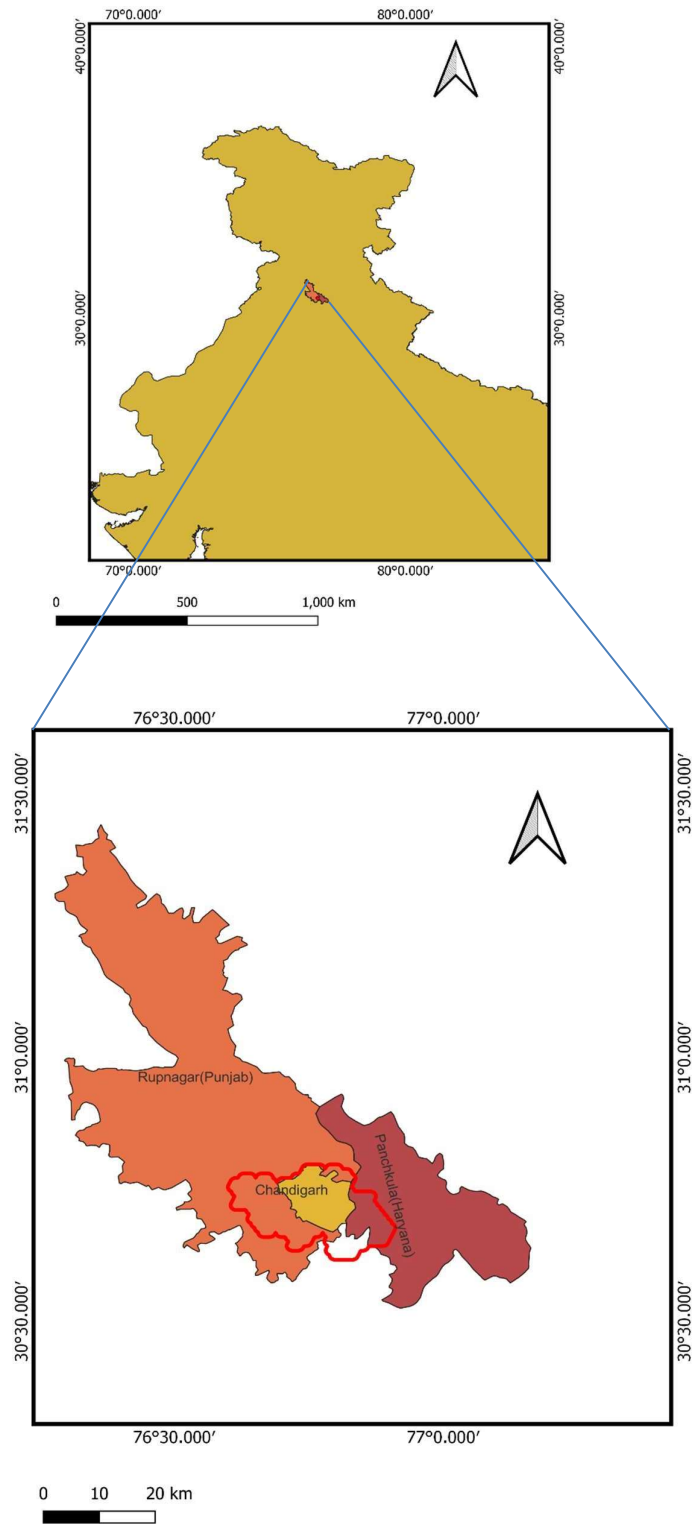


Figure 1: Area of Interest (AOI)

Chapter 3

Dataset and Methodology

The present study identifies the regions that are most exposed to the UHI effects by analyzing the spatial and temporal changes in the thermal environment of these cities with remote sensing data. The remote sensing datasets used in the present study are ready-to-use datasets retrieved from the Google Earth Engine (Gorelick et al., 2017). The flowchart of the methodology is provided in Fig. 2.

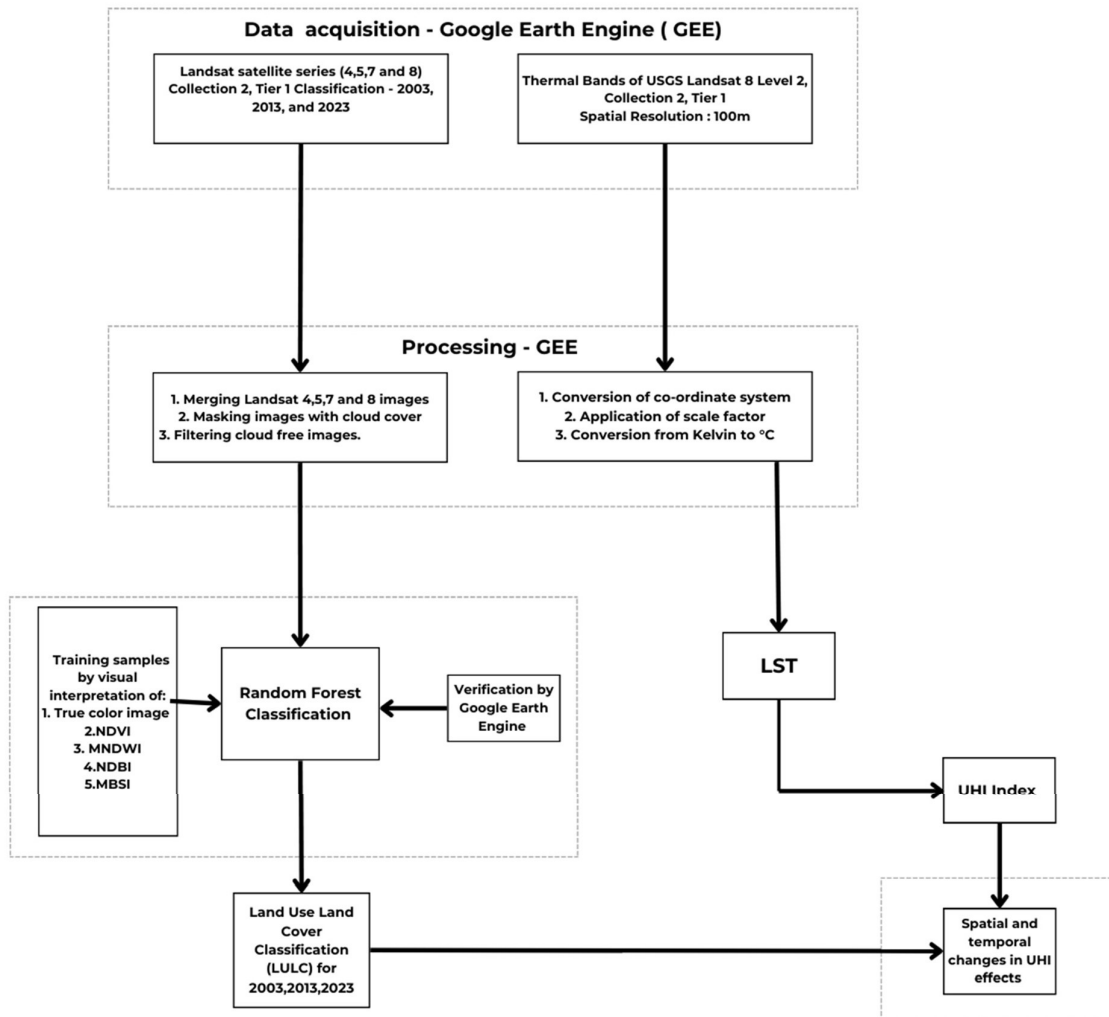


Figure 2: Flowchart

3.1 Delineation of the study area

- a. Human Settlement Layer for Chandigarh region extracted from GHS Urban Centre Database 2015, Public Release 2019, Version 1.2
- b. Region from the states of Punjab(Rupnagar), Haryana(Panchkula) and Chandigarh UT
- c. Reprojected to EPSG:32644 - WGS 84 / UTM zone 44N.
- d. Buffered for 1km to ensure all agglomerations are included.
- e. Our Area Of Interest (AOI) is delineated by the marked red region.

3.2 Land cover

The multispectral Landsat series and Sentinel-2A Level-2 (Landsat 7, and 8) satellite data (cloud-free images) acquired on annual basis of 2003, 2013, and 2023 is used for the land cover classification of the study areas. The smileRandomForest machine learning classifier in Google Earth Engine was used to classify the study area (Amani et al., 2019; Souza et al., 2020). The visual interpretation of the true color image and additional supplemental data (spectral indices) such as the NDVI (Szabo et al., 2016), the Modified Normalized Difference Water Index (MNDWI) (Szabo et al., 2016), the Normalized Difference Built-up Index (NDBI) (Zha et al., 2003), and the Modified Bare Soil Index (MBSI) (Nguyen et al., 2021) were used to provide sample points for training and testing the classifier. The inclusion of various spectral indices (NDVI, MNDWI, NDBI, and MBSI) helps to improve the accuracy of LULC classification for machine learning models (Kavhu et al., 2021). The study areas are classified into four broad land cover types: water, vegetation, built-up, and bare soil.

SENTINAL – 2A MSI

Band	Spectral region (nm)	Wavelength range (nm)	Resolution (m)
B2	Blue	458 - 523	10
B3	Green peak	543 - 578	10
B4	Red	650 - 680	10
B8	NIR	785 - 899	10
B11	SWIR	1565 - 1655	20

Table 1: Sentinel – 2A MSI Bands

LANDSAT 8 OPERATIONAL LAND IMAGER (OLI) AND THERMAL INFRARED SENSOR (TIRS) – LAUNCHED ON FEBRUARY 11, 2013

Bands	Wavelength (micrometers)	Resolution (meters)
Band 10 – Thermal Infrared (TIRS) 1	10.60 – 11.19	100
Band 11 – Thermal Infrared (TIRS) 2	11.50 – 12.51	100

Table 2: Landsat 8 OLI TIRS Bands

3.2.1 Random Forest classifier

A random forest classifier is widely used for the classification of land cover using remote sensing data (e.g., Jin et al., 2018; Maxwell et al., 2019; Phan et al., 2020). A random forest classifier is made up of several decision trees, as shown in Fig. 3. The classifier is provided with training samples by the user with specified classes, from which the trees learn different characteristics of the class. When classification is carried out, each tree casts a unit vote for the most probable class.

3.3 Land Surface Temperature

Pre-processed datasets containing atmospherically corrected surface reflectance and land surface temperature derived from the data produced by the Landsat 4,5,7,8 was used for obtained the LST of the study area for the temporal range from 2003 to 2013 to 2023. The conversion from Kelvin to °C are done in Google Earth Engine. Refer code snippet in **Annexure** for further reference.

3.4 Urban Heat Island Index

The changes in UHI are derived using the LST pattern. The UHIindex is determined using the following equation:

$$UHI_{Index} = \frac{(LST - LST_{min})}{(LST_{max} - LST_{min})}$$

where LST_i is the value of LST in the image, and LST_{max} and LST_{min} are the maximum and minimum LST values, respectively.

3.5 Thermal Classification through Hot Spot Analysis

A crucial aspect of this research involves classifying the study area based on temperature variations. To achieve this, we leverage hotspot analysis, a spatial statistical tool within ArcMap 10.8.2. The administrative boundary of the study area serves as the base layer, while normalized Land Surface Temperature (LST) data acts as the primary input for classification.

To identify statistically significant patterns of temperature clustering, we employ the Global Moran's I statistic calculated in ArcMap. This analysis considers each 30-meter squared area within the study region. It acknowledges that neighboring areas influence the thermal classification of a specific target area. The classification prioritizes the influence of nearby neighbors compared to distant ones. In this study, eight neighboring areas are factored in for each target area during classification.

The hotspot analysis categorizes the study area into eight distinct classes based on

LST and the presence of spatial clustering. These classifications include Hot Spots with varying confidence levels (99%, 95%, and 90%) signifying statistically significant clusters of high temperatures. Areas with moderate temperature ranges and no significant clustering are designated as Non-Significant. Finally, Cold Spots with varying confidence levels represent statistically significant clusters of low temperatures within the study area.

This thermal classification is instrumental in identifying highly vulnerable areas. By pinpointing zones experiencing the most intense heat (high-confidence Hot Spots), the study allows us to document the potential vulnerability of populations residing in these regions compared to others. This information provides a critical foundation for developing targeted mitigation strategies to protect residents from the adverse effects of Urban Heat Islands (UHIs).

3.6 Google Earth Engine

Google Earth Engine (GEE) stands as a transformative platform that revolutionizes the way we analyze and visualize geospatial data on a global scale. Developed by Google, this cloud-based platform provides a vast repository of Earth observation data, allowing users to access and analyze petabytes of satellite imagery and environmental datasets. With its powerful computational capabilities, Google Earth Engine enables the processing of large-scale geospatial data analyses without the need for extensive computing resources. Its user-friendly interface and extensive library of pre-processed datasets make it a valuable tool for accelerating research and decision-making processes, fostering collaboration, and addressing. You can find our project GEE repository link in **Annexure**.

Chapter 4

Result

4.1 Land cover change

The spatial changes in the land cover of Chandigarh over the past two decades show that there is a decrease in water bodies, and soil cover throughout the study area and an increase in the built up area. Chandigarh had a built-up area of 148 km² in 2003, which increased to 189 km² (127.28%) by 2013 and to 213 km² (143.91%) by 2023.

Table 3a: LULC 2003

Class	Urban	Bare land	Water	Vegetation
area	148.255	155.462	1.913	75.619
%	38.88666	40.77702	0.501772	19.83454

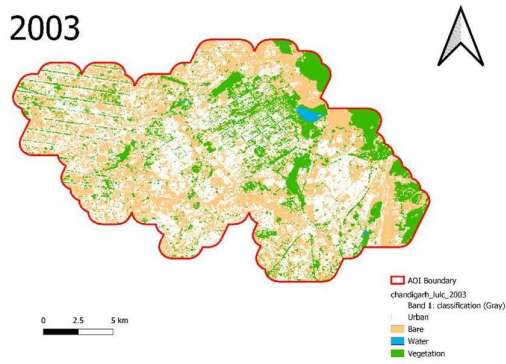
Table 3b: LULC 2013

Class	Urban	Bare land	Water	Vegetation
area	188.85	100.251	1.723	90.425
%	49.53456	26.29541	0.451936	23.71809

Table 3c: LULC 2023

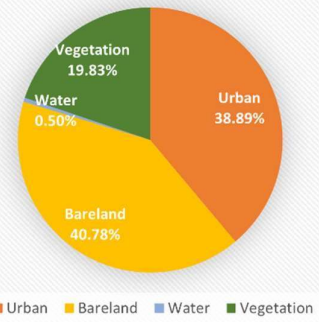
Class	Urban	Bare land	Water	Vegetation
area	213.020	38.76234	1.677	127.44858
%	55.8743	10.16602	0.43987	33.5242

Table 3: LULC data for 2003, 2013, 2023

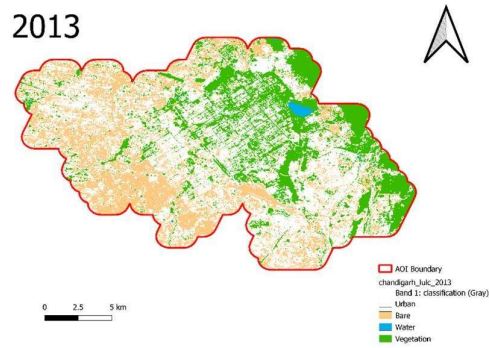


(a)

LandUse Land Cover 2003

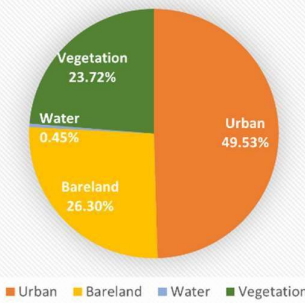


(d)

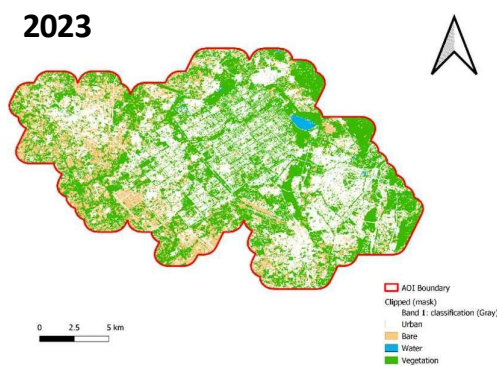


(b)

LandUse Land Cover 2013

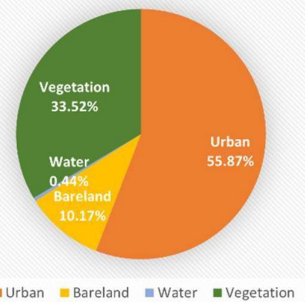


(e)



(c)

LandUse Land Cover 2023



(f)

Figure 3: LULC Classification Results

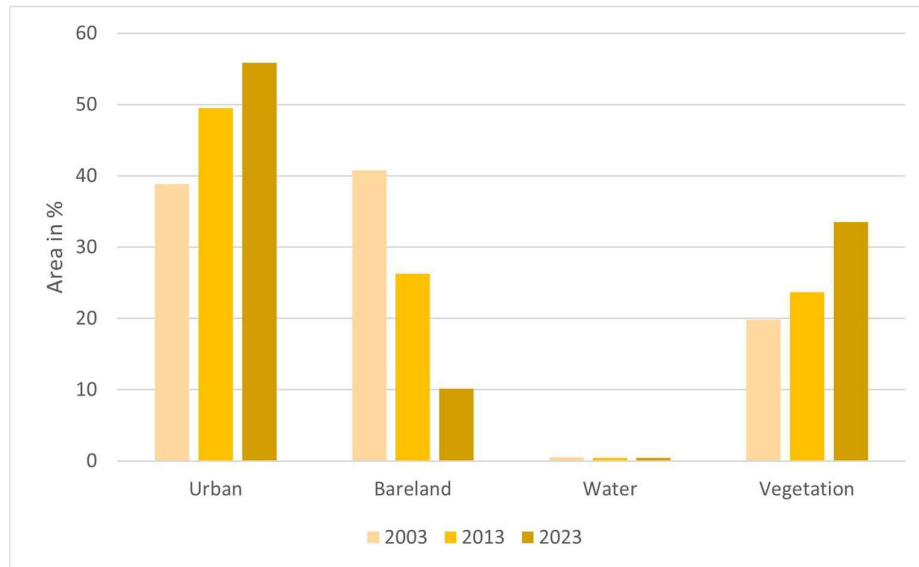
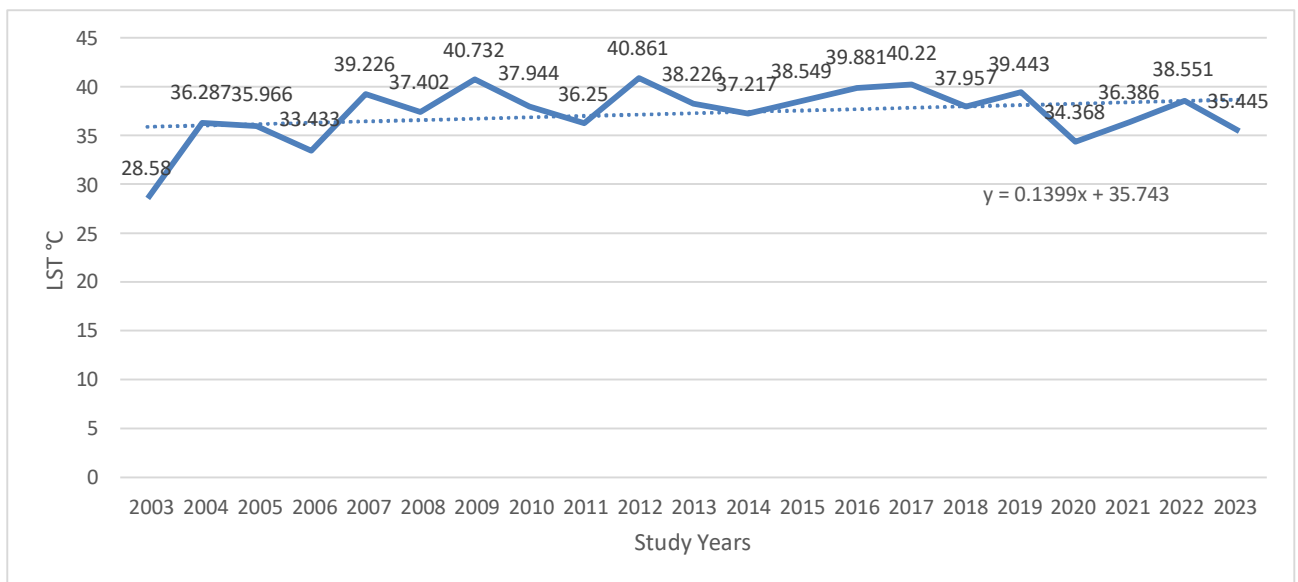


Figure 4: Comparative Graph for LULC

4.2 Land Surface Temperature

For Chandigarh, the average LST increasing by 1.538°C in the first decade (2003-2013) and 1.399°C in the second decade (2013-2023). Over the past two decades, the average LST of urbanized areas has risen by 2.789°C, demonstrating the substantial warming effect of urban development. Whereas for the unplanned city of Ludhiana it is 3.7235 in the first decade (2003-2013) and 3.385°C in the second decade (2013-2023).



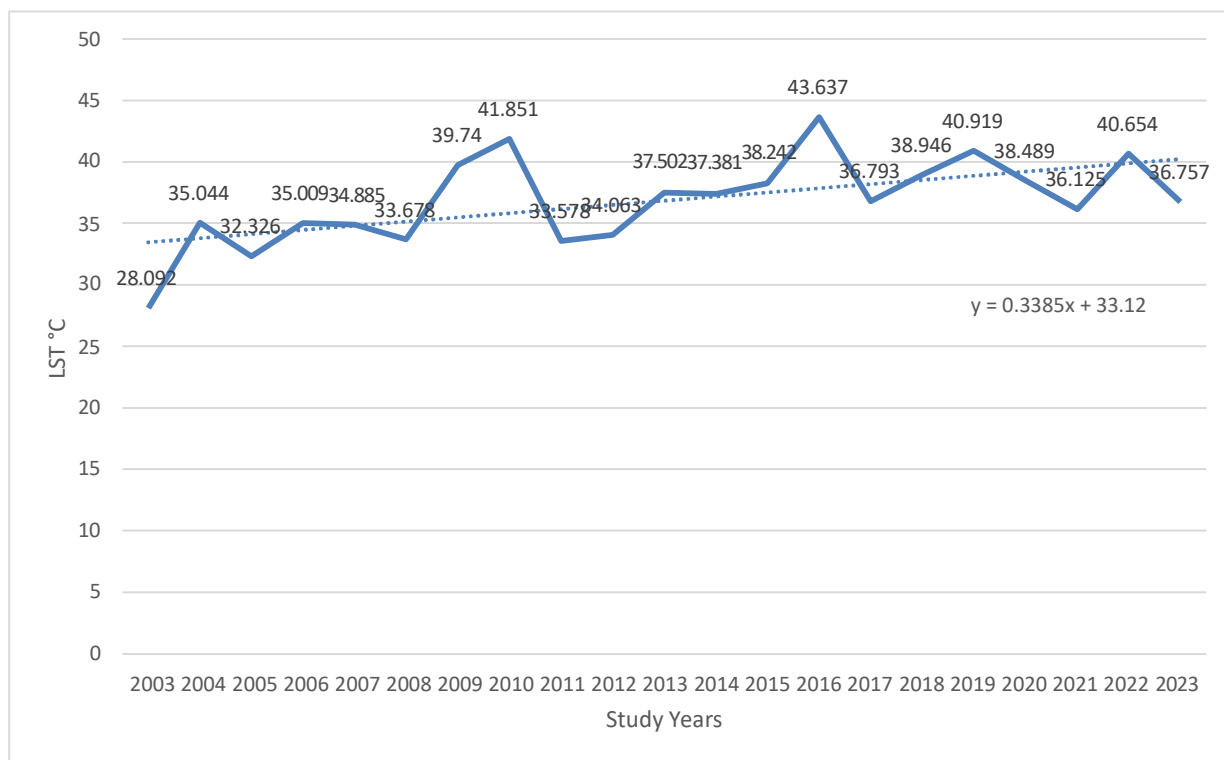


Figure 6: Average LST for Summer Months for Ludhiana

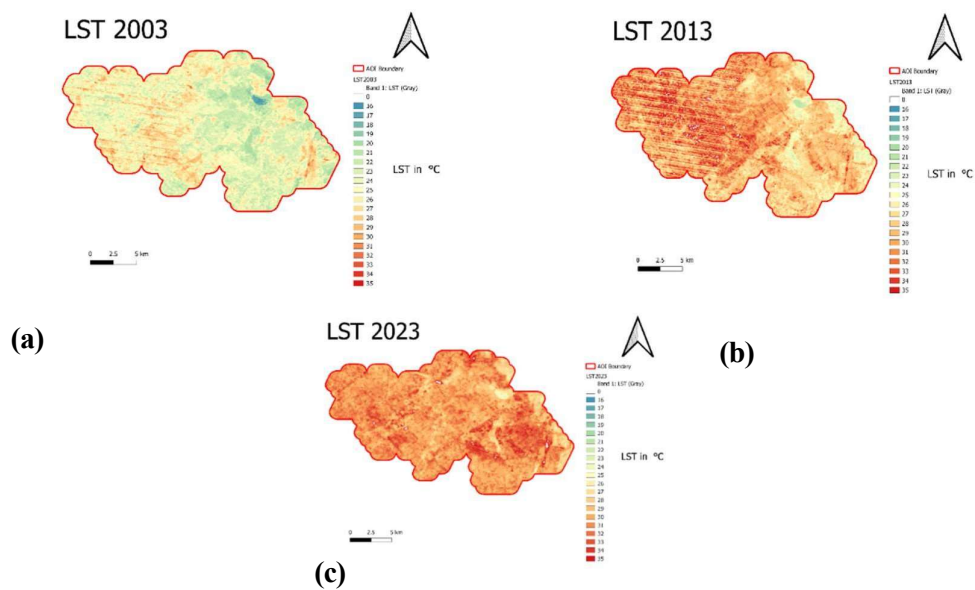


Figure 7: LST Visualizations for Chandigarh

4.3 Land Surface Temperature Time-Series (Landsat Collection 2)[2013-2022]

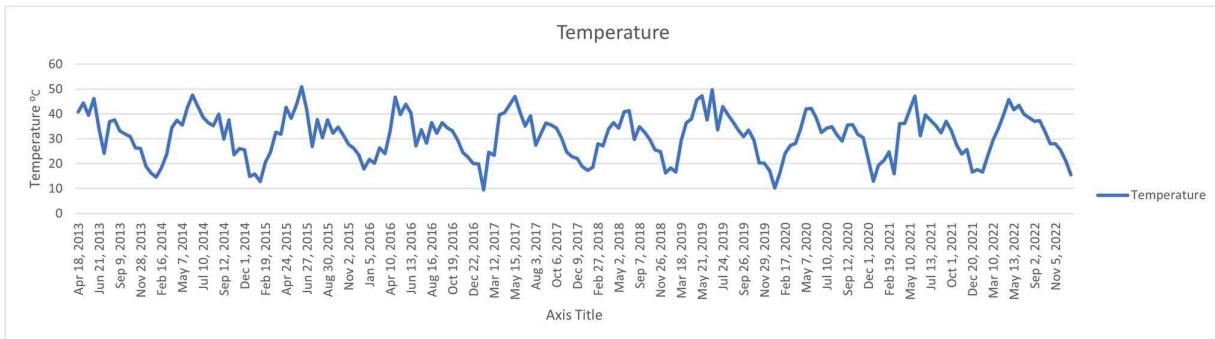
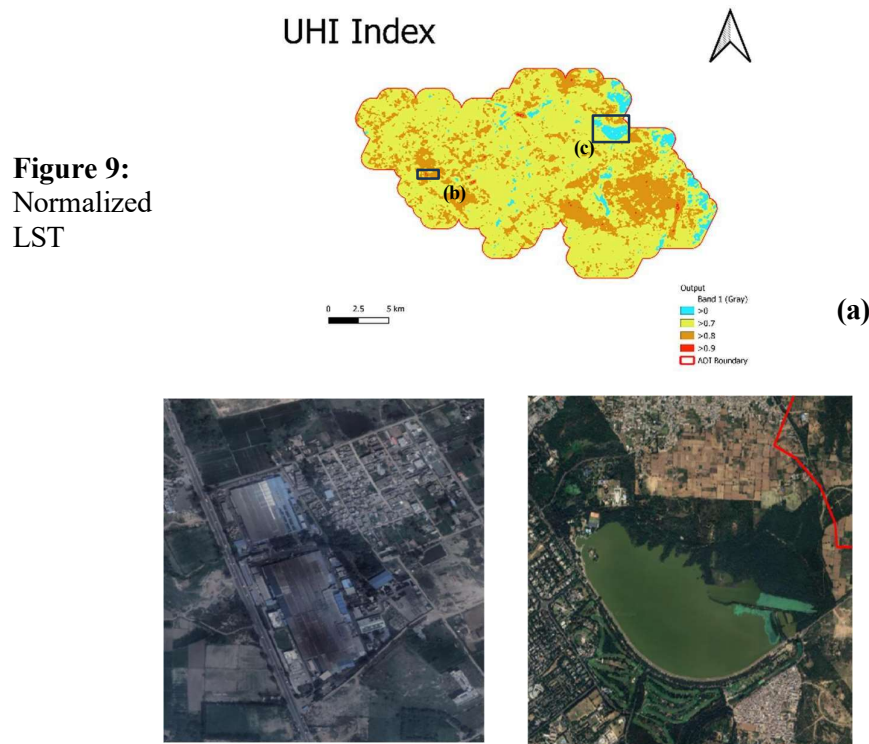


Figure 8: LST Time-Series [2013-2022]

4.4 UHI Index

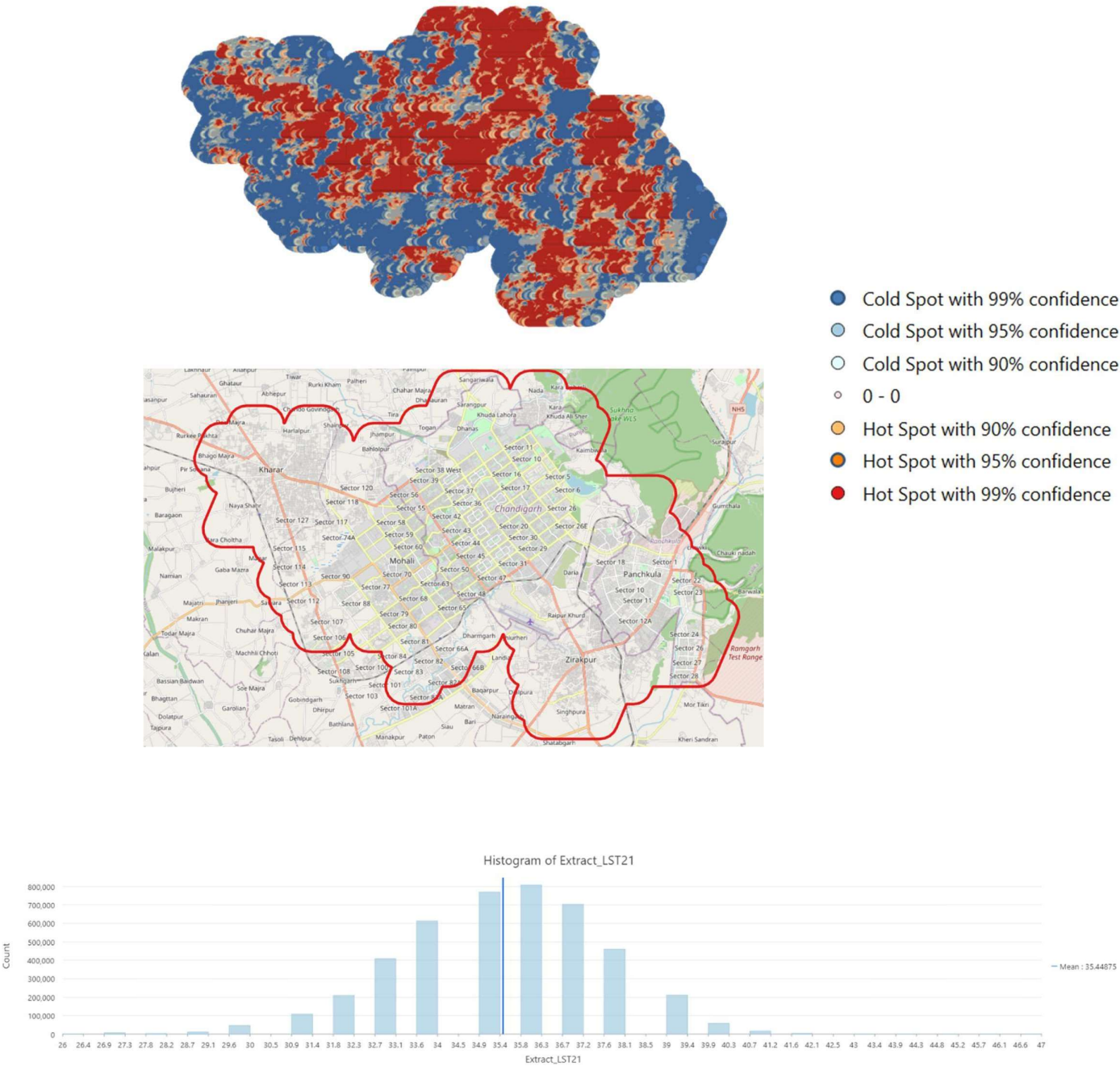
UHI Index is basically the normalized LST. The lower average UHI values are found in a large water body (Sukhna Lake) present on the north-eastern side of the city, indicating no significant heating. However, the average UHI values increased in settlement areas where sparse settlement areas have been converted into dense settlement areas. Due to the presence of industries, the southern side of the city possessed a strong UHI zone. The average UHI map depicts the areas associated with higher or lower UHI zones created throughout the city.



4.1 Thermal Classification

Spatial Autocorrelation was also performed on the dataset and Global Moran's I was calculated to be 0.988604, indicating more clustering tendency and not randomly as indicated by the z-score.

Figure 10: Hotspot Analysis result



Chapter 5

Conclusion

This study employed spatial analysis of Landsat-derived Land Surface Temperature (LST) data to comprehensively assess the Urban Heat Island (UHI) effect in Chandigarh, India, over the past two decades (2003-2023). Our findings reveal a concerning trend of rising LST across the city, with a significant increase in the spatial extent of high-temperature zones. This observed rise in LST surpasses the average values reported in global and Asian studies, highlighting the urgency of addressing UHI in Chandigarh.

A critical insight from this research is the uneven distribution of the UHI effect within Chandigarh. This highlights the well-documented impact of urbanization on local climate, where built environments contribute to heat trapping and re-radiation.

Interestingly, this study also compared Chandigarh, a planned city, to Jalandhar, an unplanned city in the region. The analysis revealed a significantly higher average increase in LST for Jalandhar (0.3385°C per year) compared to Chandigarh (0.1399°C per year) over the same two-decade period. This suggests that Chandigarh's planned layout, with designated green spaces and potentially more reflective building materials, may offer some mitigation against the UHI effect compared to unplanned urban sprawl.

The uneven distribution of the UHI effect necessitates region-specific policies to ensure the well-being of the population, particularly in densely populated areas. Future research efforts could involve investigating the social and health vulnerabilities associated with spatial variations in UHI intensity.

Looking ahead, as India ushers in a wave of metropolitan city development projects like the Delhi Mumbai Industrial Corridor (DMIC) and Smart Cities Mission, proactive measures to mitigate UHIs are essential. The learnings from Chandigarh can serve as a blueprint for other Indian cities. Large-scale green infrastructure projects, promotion of cool materials, sustainable building practices, and innovative urban design that prioritizes walkability and passive cooling are all crucial practices that can be integrated alongside development plans.

Additionally, exploring the potential impact of climate change scenarios on UHI dynamics within Chandigarh would provide valuable insights for long-term urban planning strategies.

Chapter 6

Discussions

Over the past two decades, Chandigarh experienced exponential growth in the built-up area, with an average annual growth rate of 2.19%. The conversion of the various natural land surfaces (vegetation cover, water bodies, and bare soil) to built-up areas resulted in an increase in the average LST of these areas by 6.049°C in Chandigarh. A spatial growth in the high-temperature regions of Chandigarh is also observed from the LST and normalized LST images. However, shrinkage in the size of the low-temperature region in Chandigarh was not apparent in the LST image, whereas it was clearly identifiable in the normalized LST image. Urban and regional planners in numerous cities often overlook the spatial positioning of the most vulnerable populations when devising climate mitigation plans (Johnson et al., 2009). It is crucial to customize risk assessment systems to account for the overlap between high-temperature areas and the population density in order to efficiently prioritize mitigation strategies.

When we calculated the normalized LST, the classes are divided according to the raster histogram as shown in the figure below. Maximum values are in the range of 0.6 to 0.9. This helps us to visualize the UHI effect with more feasibility, if we had divided the pixels in quantile or equal interval a particular color shade have dominated the map.

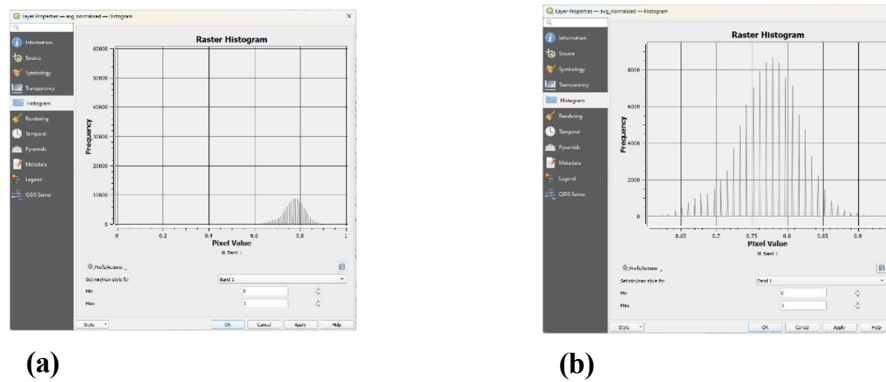


Figure 11: Raster Histogram of Normalised LST

Interestingly, our preliminary comparison of Chandigarh with Jalandhar, an unplanned city in the region, revealed a potentially significant finding. Although a detailed analysis of Jalandhar's UHI dynamics is beyond the scope of this current study, the initial data suggests that Chandigarh, with its designated green spaces,

may experience a slower rise in LST compared to unplanned cities. This finding offers a glimmer of hope and underscores the importance of thoughtful urban design in mitigating UHIs.

Urbanization in Chandigarh not only alters land cover but also disrupts the natural water cycle. Increased construction and reduced green spaces lead to decreased soil infiltration and increased surface runoff. This, in turn, causes the water table and groundwater levels to decline. With less available water, evapotranspiration, a crucial process for regulating surface temperature through evaporation from plants and soil, is hindered. Climate change is likely to exacerbate this situation, leading to a further deterioration of the water balance in Chandigarh.

Land use changes in Chandigarh can also significantly impact the local climate beyond the loss of vegetation. The conversion of agricultural land with high albedo (reflectivity) to built-up areas affects surface albedo and disrupts the balance of Earth's radiation. While the impact of vegetation loss on LST may be less pronounced compared to urban water bodies and forests, it still contributes to a decline in the city's overall natural cooling capacity. The proportional area of different land cover types within Chandigarh, including vegetation, water bodies, and built-up areas, plays a critical role in determining the overall impact on LST.

Evaluating and mapping the social impacts of Urban Heat Islands (UHI) can play a crucial role in increasing awareness among policymakers, planners, and the general public. This heightened awareness can foster support for initiatives and policies aimed at alleviating the detrimental consequences of UHI. Integrating thermal vulnerability zones and UHI social impacts into spatial planning policies enables cities to strive for improved quality of life for residents, decreased energy consumption, and effective mitigation of adverse UHI effects. Enhancing comprehension of Urban Heat Island (UHI) human health implications involves integrating demographic and health data for vulnerable populations.

Our research opens doors for further investigations. Future studies could delve deeper into the social and health vulnerabilities associated with spatial variations in UHI intensity. Certain populations, like the elderly and low-income residents who may have limited access to air conditioning, are likely to be disproportionately impacted by extreme heat events. Understanding these social vulnerabilities is crucial for developing targeted interventions to protect these populations.

Additionally, exploring the potential impact of climate change scenarios on UHI dynamics within Indian cities is essential for long-term urban planning strategies.

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Annexure

Urban Heat Island Effect Analysis of Chandigarh City on Google Earth Engine

Final Code for Outputs and Results

Below is the code repository link and QR which was utilized for conducting the analysis of the Urban Heat Island effect in Chandigarh City using Google Earth Engine:

https://code.earthengine.google.co.in/?accept_repo=users/Kartikeya/FinalYearProject



The analysis conducted revealed significant insights into the Intra-City Urban Heat Island effect in Chandigarh City. This information can be used to prioritize interventions and resources to protect residents from the harmful effects of the urban heat island.